# Natural Language Processing Course

# Project Description

## Project Description

The objective of this project is to design, implement, evaluate, and demonstrate an **end-to-end Natural Language Processing (NLP) system** that addresses a real-world language problem. Each team must formulate an NLP problem that can be treated as a meaningful computational task.

The project must answer the following question:

**How can NLP techniques be used to solve a real-world language problem, and which approach provides the best solution under realistic constraints?**

The solution should not simply demonstrate that a model works. Students must investigate **why it works, when it fails, and how different approaches compare.**

## Project Requirements

The project must satisfy the following requirements:

1. **Problem Definition:** Define one real-world NLP problem, at least 3 project objectives, and at least 1 research question.

2. **Dataset**: Use at least 5,000 text instances, unless a smaller dataset is justified. Document the data source, attributes, labels, and collection methodology.

3. **Exploratory Data Analysis:** Provide at least 4 visualizations or statistical analyses covering text length, data quality, class distribution, vocabulary, or other relevant characteristics.

4. **Text Preprocessing:** Implement at least 4 applicable preprocessing operations, such as tokenization, normalization, stop-word removal, stemming, or lemmatization.

5. **Text Representation**: Implement and compare at least 4 text representation techniques, including at least 2 traditional representation (e.g., TF-IDF) and 2 embedding-based representation.

6. **Model Development**: Implement at least 3 models or distinct configurations, including a baseline and at least 1 advanced approach appropriate to the selected task.

7**. Experimental Evaluation**: Conduct at least 4 experiments investigating model performance, preprocessing, representation, hyperparameters, or another relevant factor.

8. **Performance Metrics**: Report at least 4 relevant evaluation metrics for example classification tasks: accuracy, precision, recall, and F1-score.

9. **Comparative Analysis**: Present at least 1 comparison table covering all implemented models or configurations and identify the best-performing approach.

10. **Error Analysis**: Analyze at least 20 incorrect predictions or failure cases and categorize the major error types, where the dataset and task permit.

## Project Deliverables:

1. Detailed problem definition
2. Dataset preparation pipeline
3. NLP pipeline
4. Models Evaluation
5. Running Demo
6. GitHub Repo
7. Final Project Presentation

## Project Timeline

| # | Task Description | Deadline |
|---|---|---|
| **1** | **1. Groups Formulation**<br>2. **Idea Description** | **Week 4** |
| **2** | Dataset preparation pipeline | Week 6 |
| **3** | NLP Pipeline | Week 7 |
| **4** | Models Implementation | Week 9 |
| **5** | Final Project presentation | TBD |

## Group Formation Requirements

1. Students must form teams of 2–3 members.
2. Each team must select one NLP project topic from the suggested list or propose an alternative for instructor approval.
3. All team members must contribute to the implementation, documentation, and final presentation.
4. Team composition and project topics must be submitted for instructor approval by Week 4.
5. Each team must submit a brief description of individual contributions with the final project.
6. Individual members will be assessed **separately** during the final presentation and technical Q&A.

   Registration Form: https://forms.cloud.microsoft/r/mMQ2gY7avz?origin=lprLink